

MD. ZAWADUL ISLAM

Electrical & Electronic Engineering Undergraduate

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SUMMARY

Electrical & Electronic Engineering undergraduate at Ahsanullah University of Science and Technology with hands-on experience across power systems, protection, embedded systems, robotics, digital signal processing and electronics. Experienced in technical problem solving, project execution, team coordination and engineering communication through academic projects and leadership roles. Interested in engineering, technology, energy, operations, project management and multidisciplinary roles within dynamic organizations.

EDUCATION

Ahsanullah University of Science and Technology (AUST)

Sep 2022 – Dec 2026

BSc in Electrical & Electronic Engineering; CGPA: 3.636

Chittagong Cantonment Public College

Jun 2019 - Dec 2021

Higher Secondary Certificate; GPA: 5.00

Halishahar Cantonment Public College

2017 - 2019

Secondary School Certificate; GPA: 5.00

LEADERSHIP & ORGANIZATIONAL EXPERIENCE

Vice President, AUST Robotics Club

Jan 2026 – July 2026

- Lead technical and organizational activities involving robotics, embedded systems, workshops and engineering events.
- Coordinate project teams and support planning, implementation and technical execution of engineering activities.
- Supervise cross-functional collaboration among technical, organizing, design and promotional teams.
- Support event planning, participant engagement, technical communication and team coordination.
- Contribute to engineering competitions and large-scale student technical initiatives.

Organizing Director, AIAA AUST Student Branch

Dec 2024 - Jun 2025

- Planned and executed engineering seminars, workshops and technical events.
- Coordinated organizing, technical, design and promotional teams.
- Managed participant communication, registration and event operations.
- Supported task distribution, timelines, logistics and cross-functional coordination.

Campus Partner, Ulterior Engineering Ltd.

2024 - 2025

- Represented the organization within the university community.
- Supported communication and engagement between the organization and university students.
- Promoted engineering opportunities, initiatives and student-focused activities.

Brand Representative, Interactive Cares

2024 - 2025

- Represented Interactive Cares within the university community.
- Promoted educational programs and student-focused opportunities.
- Supported student engagement, communication and campus-level outreach activities.

SELECTED ENGINEERING PROJECTS

Overload Detection & Protection System - Arduino /

Power Systems

- Built an ACS712-based current monitoring and protection system with a 700 mA trip threshold, dual-relay load isolation, buzzer indication, and I2C LCD monitoring.
- Validated normal and overload operation experimentally: 0.43 A measured for one 100 W bulb and 0.88 A for two bulbs, with successful protective tripping.

Obstacle Detecting & Object Handling Bot - Arduino

- Developed an autonomous mobile robot using Arduino Uno, HC-SR04 ultrasonic sensing, L298N motor control, and servo-based robotic arm/gripper operation.
- Led Arduino programming and contributed substantially to physical bot integration, presentation preparation, and demonstration.

AR Frequency Estimation of Voice Signal - MATLAB (DSP)

- Implemented Burg, Yule-Walker, and Levinson-Durbin autoregressive methods to model a recorded voice signal.
- Estimated the first seven dominant frequencies and compared frequency-estimation behavior and model performance.

Digital Voting System Using Sequential Logic Circuits

- Designed a real-time vote counting and display system using 74LS90 counters, 74LS47 BCD-to-seven-segment decoders, and 74HC85 comparator logic.
- Implemented debounced vote inputs, reset functionality, seven-segment tally display, and real-time highest-vote comparison.

AM Receiver Design & Implementation

- Applied communication-theory concepts to AM receiver hardware, tuning, noise-mitigation considerations, testing, and troubleshooting in an open-ended laboratory project.

SKILLS

Engineering & Technical:

Power Systems, Power System Protection, Circuit Analysis & Design, Embedded Systems, Robotics, Digital Electronics, Signal Processing, Measurement & Instrumentation

Programming & Engineering Tools:

MATLAB, Arduino Programming, Embedded C, Proteus, Cadence Virtuoso, DipTrace, COMSOL, AutoCAD, Tinkercad

Analytical & Business:

Problem Solving, Engineering Analysis, Data Analysis, Technical Troubleshooting, Research, MS Excel, MS Office, Presentation, Technical Documentation

Leadership & Collaboration:

Technical Leadership, Project Coordination, Team Management, Event Management, Cross-Functional Collaboration, Communication, Stakeholder Engagement

ADDITIONAL INFORMATION

Languages: English, Bengali, Hindi, Urdu

Awards / Activities:

- RoboMania 2.0 – Lead.
- AUST Rover Challenge 2.0 - Segment Co-Lead.
- Robo Soccer, AUST CSE Carnival - 1st Runner-up.

Relevant Coursework: Power System Protection, Power System II, Digital Signal Processing, Communication Theory, Microprocessor Unit Lab, Digital Electronics, Measurement & Instrumentation, Electrical Machines